Chapter 5. The Tychonoff Theorem

Compactness of products, Stone–Čech compactification

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5.1 The Tychonoff theorem

Motivation. Products of compact spaces are compact, even infinite products. This is the key fact behind many existence results in analysis (for example compactness of the unit ball in the weak* topology). The proof requires Zorn's lemma, and in fact Tychonoff's theorem is equivalent to the axiom of choice.

We prove compactness via the finite intersection property (Theorem 3.23): a space is compact iff every collection $\mathcal{A}$ of subsets with the finite intersection property (FIP) satisfies $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Lemma 5.1 (37.1)

Let $\mathcal{A}$ be a collection of subsets of X with the FIP. Then there is a collection $\mathcal{D} \supset \mathcal{A}$ with the FIP such that no collection properly containing $\mathcal{D}$ has the FIP ($\mathcal{D}$ is *maximal* with respect to the FIP).

Proof. Let $\mathbb{A}$ be the set of all collections $\mathcal{B} \supset \mathcal{A}$ of subsets of X with the FIP, strictly partially ordered by proper inclusion. If $\mathbb{B} \subset \mathbb{A}$ is a chain, $\mathcal{C} = \bigcup_{\mathcal{B}\in\mathbb{B}} \mathcal{B}$ has the FIP: finitely many $C_1, \ldots, C_n \in \mathcal{C}$ lie in $\mathcal{B}_1, \ldots, \mathcal{B}_n$, all contained in the largest $\mathcal{B}_k$ (chain), which has the FIP. So $\mathcal{C}$ is an upper bound, and Zorn's lemma (Theorem 1.29) gives a maximal element. ■

Lemma 5.2 (37.2)

Let $\mathcal{D}$ be maximal with respect to the FIP.
(a) Every finite intersection of elements of $\mathcal{D}$ is in $\mathcal{D}$.
(b) If $A \subset X$ intersects every element of $\mathcal{D}$, then $A \in \mathcal{D}$.

Proof. (a) Let B be a finite intersection of elements of $\mathcal{D}$; $\mathcal{E} = \mathcal{D} \cup \{B\}$ has the FIP (a finite intersection of members of $\mathcal{E}$ is a finite intersection of members of $\mathcal{D}$), so $\mathcal{E} = \mathcal{D}$ by maximality.

(b) $\mathcal{E} = \mathcal{D} \cup \{A\}$: a finite intersection $A \cap D_1 \cap \cdots \cap D_n$ is nonempty since $D_1 \cap \cdots \cap D_n \in \mathcal{D}$ by (a) and A meets it. So $A \in \mathcal{D}$. ■

Theorem 5.3 (37.3, Tychonoff theorem)

An arbitrary product of compact spaces is compact in the product topology.

Proof. Let $X = \prod_{\alpha\in J} X_\alpha$ and $\mathcal{A}$ a collection of subsets of X with the FIP; extend it to a maximal $\mathcal{D}$ (Lemma 5.1). It suffices to find $x \in \bigcap_{D\in\mathcal{D}} \overline{D}$. For each α, $\{\pi_\alpha(D) : D \in \mathcal{D}\}$ has the FIP in the compact X_α, so choose $x_\alpha \in \bigcap_D \overline{\pi_\alpha(D)}$, and let $x = (x_\alpha)$.

Step 1. if U_β is a neighbourhood of x_β, then $\pi_\beta^{-1}(U_\beta)$ meets every $D \in \mathcal{D}$ (since U_β meets $\pi_\beta(D)$, as $x_\beta \in \overline{\pi_\beta(D)}$), so $\pi_\beta^{-1}(U_\beta) \in \mathcal{D}$ by Lemma 5.2(b).

Step 2. every basis element containing x is a finite intersection of such subbasis elements, hence lies in $\mathcal{D}$ (Lemma 5.2(a)), hence meets every element of $\mathcal{D}$ (FIP). So $x \in \overline{D}$ for all $D \in \mathcal{D}$. ■

The theorem fails for the box topology ($[0,1]^\omega$ in the box topology is not compact), and it is equivalent to the axiom of choice.

Example 5.1

$[0,1]^\omega$ is a compact metrizable space; every regular second countable space imbeds in it (proof of Theorem 4.12). $[0,1]^J$ for uncountable J is compact Hausdorff but not metrizable: it is not even first countable (a countable family of basis neighbourhoods of a point restricts only

countably many coordinates).

5.2 The Stone–Čech compactification

Motivation. Every completely regular space can be compactified, and there is a largest compactification βX in which every bounded continuous function on X extends. It is constructed by embedding X into a product of intervals and taking the closure, so Tychonoff's theorem supplies compactness.

A **compactification** of X is a compact Hausdorff space Y containing X as a subspace with $\overline{X} = Y$; two are **equivalent** if there is a homeomorphism between them that is the identity on X. If X has a compactification, it is completely regular (a subspace of a compact Hausdorff, hence normal, space).

Lemma 5.4 (38.1)

If $h : X \to Z$ is an imbedding in a compact Hausdorff space Z, there is a compactification Y of X and an imbedding $H : Y \to Z$ equal to h on X; Y is unique up to equivalence (Y is the compactification **induced** by h).

Proof. Let $X_0 = h(X)$ and $Y_0 = \overline{X_0}$, a compact Hausdorff space. Form $Y = X \sqcup (Y_0 - X_0)$ with the bijection $H : Y \to Y_0$ (h on X, identity elsewhere) and give Y the topology making H a homeomorphism. Then X is a subspace of Y, dense since X_0 is dense in Y_0. ■

Theorem 5.5 (38.2)

Every completely regular space X has a compactification Y such that every bounded continuous $f : X \to \mathbb{R}$ extends uniquely to a continuous $Y \to \mathbb{R}$.

Proof. Let $\{f_\alpha\}_{\alpha \in J}$ be all bounded continuous real functions on X, with $f_\alpha(X) \subset I_\alpha$, where $I_\alpha = [\inf f_\alpha, \sup f_\alpha]$. $h = (f_\alpha) : X \to Z = \prod I_\alpha$ is an imbedding (Theorem 4.13, using complete regularity), and Z is compact Hausdorff (Tychonoff). Let Y be the induced compactification with $H : Y \to Z$. For $f = f_\beta$, $\pi_\beta \circ H$ is a continuous extension. Uniqueness follows from the next lemma. ■

Lemma 5.6 (38.3)

If $A \subset X$ and $f : A \to Z$ is continuous with Z Hausdorff, there is at most one continuous extension $g : \overline{A} \to Z$.

Proof. If $g \neq g'$ at x, separate $g(x), g'(x)$ by disjoint U, U'; the neighbourhood $g^{-1}(U) \cap g'^{-1}(U')$ of x contains a point $y \in A$, but $g(y) = g'(y) = f(y)$ lies in $U \cap U'$. ■

Theorem 5.7 (38.4)

If Y is a compactification of the completely regular X with the extension property of Theorem 5.5, every continuous map f of X into a compact Hausdorff space C extends uniquely to a continuous $g : Y \to C$.

Proof. C is completely regular, so imbeds in some $[0,1]^J$ (Corollary 4.14); regard $C \subset [0,1]^J$. Each coordinate $\pi_\alpha \circ f$ extends to $g_\alpha : Y \to [0,1]$; $g = (g_\alpha) : Y \to [0,1]^J$ is continuous. Since $g(X) = f(X) \subset C$ and X is dense, $g(Y) \subset \overline{g(X)} \subset C$ (C is closed).

Uniqueness: Lemma 5.6. ■

Theorem 5.8 (38.5)

Two compactifications Y_1, Y_2 of X with the extension property of Theorem 5.5 are equivalent.

Proof. Extend the inclusion $j_2 : X \to Y_2$ to $f_2 : Y_1 \to Y_2$ and j_1 to $f_1 : Y_2 \to Y_1$ (Theorem 5.7). $f_1 \circ f_2 : Y_1 \to Y_1$ extends the identity of X, as does id_{Y_1}; by Lemma 5.6, $f_1 \circ f_2 = \mathrm{id}$. Similarly $f_2 \circ f_1 = \mathrm{id}$. ■

This compactification is the **Stone–Čech compactification** $\beta(X)$. It is characterized by the universal property of Theorem 5.7.

Example 5.2

(a) The one-point compactification of $(0,1)$ is the circle and $[0,1]$ is a two-point compactification; neither is $\beta(0,1)$, since $\sin(1/x)$ extends to $\beta(0,1)$.
(b) $\beta(\mathbb{Z}_+)$ is not sequentially compact: $\mathbb{Z}_+$ has no convergent subsequence in $\beta\mathbb{Z}_+$ (separate even and odd subsequences by extending functions on $\mathbb{Z}_+$).
(c) $\beta(S_\Omega) = \overline{S}_\Omega$, because every continuous real function on S_Ω is eventually constant.

5.3 Selected exercises

Motivation. These exercises from Munkres give the countable analogue of the finite intersection criterion, make precise in what sense $\beta(X)$ is the largest compactification, prove the fact about S_Ω used in Example 5.2(c), and show that β preserves connectedness.

Exercise 5.1 (§37, Ex. 2; Lindelöf spaces)

A collection $\mathcal{A}$ of subsets of X has the **countable intersection property** if every countable intersection of elements of $\mathcal{A}$ is nonempty. Show that X is Lindelöf iff for every collection $\mathcal{A}$ with the countable intersection property, $\bigcap_{A\in\mathcal{A}} \overline{A} \neq \emptyset$.

Solution.

($\Rightarrow$) Let X be Lindelöf, $\mathcal{A}$ with the countable intersection property, and suppose $\bigcap_{A\in\mathcal{A}} \overline{A} = \emptyset$. Then $\{X - \overline{A} : A \in \mathcal{A}\}$ is an open covering of X, so countably many $X - \overline{A_n}$ cover X, i.e. $\bigcap_n \overline{A_n} = \emptyset$. But $\bigcap_n A_n \subset \bigcap_n \overline{A_n}$ is nonempty, a contradiction.

($\Leftarrow$) Let $\mathcal{U}$ be an open covering with no countable subcovering, and $\mathcal{A} = \{X - U : U \in \mathcal{U}\}$. For $U_1, U_2, \cdots \in \mathcal{U}$, $\bigcap_n (X - U_n) = X - \bigcup_n U_n \neq \emptyset$, so $\mathcal{A}$ has the countable intersection property. Its members are closed, and $\bigcap_{U\in\mathcal{U}}(X - U) = X - \bigcup \mathcal{U} = \emptyset$, contradicting the hypothesis. So X is Lindelöf. ■

Exercise 5.2 (§38, Ex. 4; $\beta(X)$ is the largest compactification)

Let Y be any compactification of the completely regular space X. Show that there is a continuous, closed, surjective map $g : \beta(X) \to Y$ that is the identity on X.

Solution.

Step 1. The inclusion $j : X \to Y$ is a continuous map into a compact Hausdorff space, so by Theorem 5.7 it extends to a continuous $g : \beta(X) \to Y$, and $g = j$ is the identity on X.

Step 2. (g is closed.) A closed $C \subset \beta(X)$ is compact (Theorem 3.16), so $g(C)$ is compact (Theorem 3.19), hence closed in the Hausdorff space Y (Theorem 3.17).

Step 3. (g is onto.) By Step 2, $g(\beta(X))$ is closed in Y; it contains X, which is dense in Y. So $g(\beta(X)) = Y$.

A closed continuous surjection is a quotient map, so every compactification of X is equivalent to a quotient space of $\beta(X)$. ■

Exercise 5.3 (§38, Ex. 5(a); functions on S_Ω)

Show that every continuous $f : S_\Omega \to \mathbb{R}$ is eventually constant: there is $\alpha \in S_\Omega$ with $f(\beta) = f(\alpha)$ for all $\beta > \alpha$.

Solution.

Step 1. (Increasing sequences converge.) Let $\alpha_1 < \alpha_2 < \cdots$ in S_Ω. By Theorem 1.27 the set of upper bounds of $\{\alpha_n\}$ in S_Ω is nonempty; let b be its least element. If $c < b$, then c is not an upper bound, so $\alpha_N > c$ for some N, and $\alpha_n \in (c, b]$ for $n \geq N$. Since $b > \alpha_1$ is not the least element of S_Ω, every neighbourhood of b contains such an interval $(c, b]$ with $c < b$; hence $\alpha_n \to b$, and by continuity $f(\alpha_n) \to f(b)$.

Step 2. (Small oscillation eventually.) Fix $\epsilon > 0$. We claim there is α with $|f(\beta) - f(\alpha)| < \epsilon$ for all $\beta > \alpha$. If not, then for every α there is $\beta > \alpha$ with $|f(\beta) - f(\alpha)| \geq \epsilon$. Starting from any α_1, we obtain $\alpha_1 < \alpha_2 < \cdots$ with $|f(\alpha_{n+1}) - f(\alpha_n)| \geq \epsilon$ for all n. Then $(f(\alpha_n))$ is not Cauchy, contradicting Step 1.

Step 3. (Conclusion.) For each n choose α_n as in Step 2 with $\epsilon = 1/n$, and by Theorem 1.27 an upper bound α of $\{\alpha_n\}$. For $\beta > \alpha$ and each n, both β and α are $\geq \alpha_n$, so

$$|f(\beta) - f(\alpha)| \leq |f(\beta) - f(\alpha_n)| + |f(\alpha_n) - f(\alpha)| < \tfrac{2}{n}$$

(the second term is 0 if $\alpha = \alpha_n$). Letting $n \to \infty$, $f(\beta) = f(\alpha)$. ■

Exercise 5.4 (§38, Ex. 6; connectedness of $\beta(X)$)

Let X be completely regular. Show that X is connected iff $\beta(X)$ is connected.

Solution.

($\Rightarrow$) X is connected and dense in $\beta(X)$, and the closure of a connected set is connected, so $\beta(X) = \overline{X}$ is connected.

($\Leftarrow$) Suppose $X = A \cup B$ is a separation. Then $f = 0$ on A, $f = 1$ on B is continuous and bounded (A, B are open), so it extends to a continuous $g : \beta(X) \to \mathbb{R}$ (Theorem 5.5). Since X is dense, $g(\beta(X)) = g(\overline{X}) \subset \overline{g(X)} = \{0, 1\}$, and both values occur. So $g^{-1}(0)$ and $g^{-1}(1)$ form a separation of $\beta(X)$, a contradiction. ■